

CSA-0903 PROGRAMING IN JAVA

EXCEPTION HANDLING



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

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1. ArithmeticException – Student Calculator

Problem Statement:

Develop a Java program to calculate the average marks of students. The program should accept the total marks and the number of subjects from the user. If the number of subjects entered is **0**, the program should handle the exception gracefully and display an appropriate error message instead of terminating abruptly.

Requirements:

- Read total marks and number of subjects.
- Calculate and display the average.
- Handle ArithmeticException using try-catch.
- Display a user-friendly error message when division by zero occurs.

CODE:

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.println("enter marks:");
6         int marks=sc.nextInt();
7         System.out.println("enter number of subjects:");
8         int numberofsubjects=sc.nextInt();
9         try{
10            int average=marks/numberofsubjects;
11            System.out.println("Average="+average);
12        }
13        catch(ArithmeticException e){
14            System.out.println("error :number of subjects cannot divide by zero");
15            System.out.println("enter valid number of subjects");
16        }
17    }
18 }
19 }
20 }
```

OUTPUT:

```
enter marks:
100
enter number of subjects:
0
error :number of subjects cannot divide by zero
enter valid number of subjects

...Program finished with exit code 0
Press ENTER to exit console.
```

2. NullPointerException – Employee Management System

Problem Statement:

Develop a Java program for an Employee Management System. The program should create an employee object and display the employee's name. Demonstrate a situation where the employee object is **not initialized (null)** and handle the resulting exception gracefully.

Requirements:

- Create an Employee class with employee ID and name.
 - Attempt to access the employee object's data without creating the object.
- Handle NullPointerException using try-catch.
- Display an appropriate message instead of allowing the program to crash

OUTPUT:

```
1 class Employee {
2     int employeeId;
3     String name;
4     Employee(int employeeId, String name) {
5         this.employeeId = employeeId;
6         this.name = name;
7     }
8 }
9 public class Main {
10     public static void main(String[] args) {
11         Employee emp = null;
12         try {
13             System.out.println("Employee ID: " + emp.employeeId);
14             System.out.println("Employee Name: " + emp.name);
15         }
16         catch (NullPointerException e) {
17             System.out.println("Error: Employee object is not initialized.");
18             System.out.println("Please create the employee object before accessing its data.");
19         }
20     }
21 }
```

OUTPUT:

```
Error: Employee object is not initialized.  
Please create the employee object before accessing its data.  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

3. ArrayIndexOutOfBoundsException – Student Marks Processing Problem

Statement:

Develop a Java program to store the marks of **five students** in an array. Allow the user to enter the index of the student whose mark should be displayed. If the entered index is outside the valid range, handle the exception and display an informative message.

Requirements:

- Store marks in an integer array of size 5.
- Accept an index from the user.
- Display the mark corresponding to the entered index.
- Handle ArrayIndexOutOfBoundsException using try-catch.
- Inform the user about the valid index range.

CODE:

```
1 import java.util.Scanner;  
2 public class Main{  
3     public static void main(String[] args){  
4         Scanner sc=new Scanner(System.in);  
5         int[] marks={85,90,98,99,100};  
6         System.out.println("enter index:");  
7         int index=sc.nextInt();  
8         try{  
9             System.out.println("Student mark:"+marks[index]);  
10        }  
11        catch(ArrayIndexOutOfBoundsException e){  
12            System.out.println("array out of limit.");  
13        }  
14    }  
15 }
```

OUTPUT:

```
enter index:
8
array out of limit.

...Program finished with exit code 0
Press ENTER to exit console.
```

4. NumberFormatException – Customer Registration System Problem

Statement:

Develop a Java program for a Customer Registration System. The program should accept the customer's **age as a String** and convert it into an integer. If the user enters invalid data (such as alphabets or special characters), the program should handle the exception and display an appropriate error message.

Requirements:

- Read age as a String.
- Convert the String into an integer using Integer.parseInt().
- Display the customer's age if the input is valid.
- Handle NumberFormatException using try-catch.
- Display a meaningful message when invalid input is entered.

CODE:

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter customer age: ");
6         String age = sc.nextLine();
7         try {
8             int customerAge = Integer.parseInt(age);
9             System.out.println("Customer Age = " + customerAge);
10        }
11        catch (NumberFormatException e) {
12            System.out.println("Error: Invalid age!");
13            System.out.println("Please enter age using numbers only.");
14        }
15        sc.close();
16    }
17 }
```

OUTPUT:

```
Enter customer age: e3
Error: Invalid age!
Please enter age using numbers only.

...Program finished with exit code 0
Press ENTER to exit console.
```

CLASS ACTIVITY:

1. ArithmeticException

CODE:

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             System.out.println("enter number 1:");
7             int num1=sc.nextInt();
8             System.out.println("enter number 2:");
9             int num2=sc.nextInt();
10            int output=num1/num2;
11            System.out.println("result="+output);
12        }
13        catch(ArithmeticException e){
14            System.out.println("number cannot divide by zero");
15        }
16    }
17 }
```

OUTPUT:

```
enter number 1:
50
enter number 2:
0
number cannot divide by zero

...Program finished with exit code 0
Press ENTER to exit console.
```

2. NullPointerException

CODE:

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             String s=null;
7             System.out.println(s.length());
8         }
9         catch(NullPointerException e){
10            System.out.println("the given string is empty");
11        }
12    }
13 }
```

OUTPUT:

```
the given string is empty

...Program finished with exit code 0
Press ENTER to exit console.
```

3. ArrayIndexOutOfBoundsException

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             int a[]=new int[5];
7             a[10]=50;
8             System.out.println("Array elements:"+a[10]);
9         }
10        catch(ArrayIndexOutOfBoundsException e){
11            System.out.println("Array caanmot be executed.");
12        }
13    }
14 }
```

OUTPUT:

```
Array caanmot be executed.

...Program finished with exit code 0
Press ENTER to exit console.
```

4. NumberFormatException

```
1 public class Main {
2     public static void main(String[] args) {
3
4         String s = "ABC";
5
6         try {
7             int i = Integer.parseInt(s);
8             System.out.println("Integer value: " + i);
9         }
10        catch (NumberFormatException e) {
11            System.out.println("Invalid number format.");
12        }
13    }
14 }
```

OUTPUT:

```
Invalid number format.  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```